377067

Course Name: Database Management System

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

**Printed Pages: 4** **University Roll No. .....**

**End Term Examination, EVEN Semester 2023-24**
**B.Tech (CS/AIML/CCV/CSF/DA/IIoT/EC(CS)), Year: II, Semester: IV**
**BCSC1003 Database Management System**

**Time: 3 Hours** **Maximum Marks: 50**

Instruction for students: ----------

1. Attempt all questions.
2. Do not switch between the section A, Section B and Section C.
3. Use only a blue pen to answer.
4. No B sheet will be given.

**Section – A**

*Attempt All Questions* 4 X 5 = 20 Marks

| No. | Detail of Question | Marks | CO | BL | KL |
|---|---|---|---|---|---|
| 1 | Suppose you're tasked with designing a relational database for a university's course registration system. The functional dependencies (FDs) for the database schema are given as follows:<br>F = {CourseCode → CourseTitle, CourseCode → Instructor, Instructor → Department, Department → Building, Building → Room}<br>Find the minimal cover of the given set of functional dependencies. | 4 | 6 | An | P |
| 2 | The database consists of three tables:<br>**Cars:** Contains information about cars available for sale, including car ID, make, model, and price.<br>**Customers:** Contains information about customers interested in purchasing cars, including customer ID, name, email, and phone number. | 4 | 4 | An | P |

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| | **Sales:** Records each car sale, including sale ID, car ID, customer ID, and sale amount.<br>Write the following queries in relational algebra:<br>a. Find the names of customers who reside in "Los Angeles."<br>b. Find the details of cars whose make is "Toyota."<br>c. Find the customer IDs of customers who have purchased cars for more than $20,000.<br>d. Find the customer IDs of customers who have purchased cars of a certain make, let's say "Honda." | | | | |
| 3 | Imagine you're the IT manager of a multinational corporation that relies heavily on its database management system (DBMS) to handle a vast amount of data across various departments. Let's explore the significance of data independence within this scenario. | 4 | 2 | A | C |
| 4 | You've been tasked with designing a robust transaction system for a new online banking platform. The system must ensure the integrity and reliability of financial transactions. Discuss the ACID properties crucial for maintaining transactional integrity in this banking system. Provide a detailed explanation of each property and illustrate them with examples tailored to the context of online banking. | 4 | 3 | An | P |
| 5 | Given a relation **R(A, B, C, D)** with the following set of functional dependencies (FDs): **A→BC, CD→D**, the relation R is decomposed into two new relations:<br>**a).** D1:(A, B) and D2:(C, D) **b).** D1:(A,B,C) and D2:(A,D)<br>Check whether each decomposition is lossless or lossy, and justify your answer for each. | 4 | 8 | U | F |

**Section – B**

*Attempt All Questions* 3 X 5 = 15 Marks

| No. | Detail of Question | Marks | CO | BL | KL |
|---|---|---|---|---|---|
| 6 | Explain the significance of ensuring recoverability in schedules. Are there any situations where allowing non-recoverable schedules would be preferable? | 3 | 8 | C | C |

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|---|---|---|---|---|---|
| 7 | Define indexing in the context of a database management system (DBMS) and explain its significance in improving query performance. | 3 | 6 | An | P |
| 8 | Consider a database schema for an e-commerce platform with the following tables:<br>**Customers** (customer_id, customer_name, email)<br>**Orders** (order_id, customer_id, order_date, total_amount)<br>**Order_Items** (order_item_id, order_id, product_id, quantity, unit_price)<br>**Products** (product_id, product_name, category, price)<br>Write an SQL query to retrieve the names of customers who have placed at least two orders in the month of January 2024. Additionally, display the total number of orders placed by each customer in January 2024. | 3 | 7 | A | P |
| 9 | Define a distributed database and elaborate on the concept of fragments along with their types. Provide examples to illustrate each type of fragment within the context of distributed databases. | 3 | 8 | U | F |
| 10 | Consider a relation schema T with three attributes **XYZ** (listed in order). Suppose there are three tuples in a legal instance of relation T: (A, B, C), (D, B, E), and (F, G, H). Write down the functional dependency that holds over schema T. | 3 | 6 | A | C |

## Section – C

*Attempt All Questions*

5 X 3 = 15 Marks

| No. | Detail of Question | Marks | CO | BL | KL |
|---|---|---|---|---|---|
| 11 | Consider a database system using the Two-Phase Locking (2PL) protocol for concurrency control. The system has two transactions, T1 and T2, accessing a shared resource.<br>**Transaction T1** acquires an exclusive lock on the resource and performs some operations. | 5 | 7 | U | F |

| | | | | | |
|---|---|---|---|---|---|
| | **Transaction T2** requests a shared lock on the same resource.<br>**Transaction T1** releases its lock after completing its operations.<br>**Transaction T2** then acquires a shared lock and performs its operations.<br><br>Describe the behavior of the transactions **T1** and **T2** according to the Two-Phase Locking (2PL) protocol. Explain how 2PL ensures serializability and prevents conflicts between concurrent transactions. | | | | |
| 12 | Consider a database system that uses log-based recovery for ensuring data consistency and reliability. The log file for a particular transaction consists of the following entries:<br><br>Start transaction T1<br>Write item X, old value = 10, new value = 20<br>Write item Y, old value = 30, new value = 40<br>Commit transaction T1<br><br>Assuming that the system crashes just after entry 3, explain how log-based recovery would be used to restore the database to a consistent state upon system restart. Provide step-by-step instructions for the recovery process. | 5 | 8 | C | M |
| 13 | Consider the following schedule S:<br><br>**S: R1(A), W2(A), R2(B), W3(B), R1(B), W1(A)**<br><br>Check whether the given schedule S is view serializable or not. If it is view serializable, provide the serial schedule. If it is not view serializable, explain why? | 5 | 6 | An | P |